TRoss 606 User Manual

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Overview

TRoss 606 is an 8-voice, 606-style drum synthesizer with a built-in 16-step sequencer, 12 pattern slots and recording from your MIDI keyboard. It runs as an AU, VST3 or LV2 plug-in, or as a standalone app, on macOS, Windows and Linux.

Item	Detail
Voices	Bass drum, snare, low tom, hi tom, clap, closed hat, open hat, cymbal
Sequencer	16 steps, 9 rows (accent + 8 drums), length 1–16, synced to the host tempo
Patterns	12 slots, switched at the end of the playing pattern
Formats	macOS: AU, VST3, app · Windows: VST3, app · Linux: VST3, LV2, app
Output	Stereo main out, plus 8 mono outputs, one per drum
Version	0.8.0 (shown at the top right of the panel)

What's new in 0.8.0: **flams**. Right-click any drum pad to give that step a flam, drag or ruff (one, two or three grace notes just before the hit), with its own spacing and grace-note level (see *Flams*). Click the TRoss 606 logo for the new **About** window, with the contact address and the manual built into the plug-in (see *About and the built-in manual*). The clap's default Decay is now 100 %. Version 0.7 added note repeat, 0.6 pattern copy and chains, 0.5 EXPORT MIDI, 0.4 the playable name strips, 0.3 separate drum outputs, and 0.2 the hardware-style panel and 12 pattern slots.

Installation

Download the installer for your system from the latest successful run on the repository's Actions page, under **Artifacts** (you must be signed in to GitHub).

macOS

1. Unzip **TRoss606-0.8.0-macOS** and open **TRoss606-0.8.0-macOS.pkg**.
2. The installer isn't signed, so macOS blocks it the first time. Open System Settings → Privacy & Security and click **Open Anyway**.
3. Click **Customize** if you want to skip the AU, the VST3 or the standalone app. Everything else installs system-wide:
   - `/Library/Audio/Plug-Ins/Components/TRoss 606.component` (AU)
   - `/Library/Audio/Plug-Ins/VST3/TRoss 606.vst3`
   - `/Applications/TRoss 606.app`

Windows

1. Unzip **TRoss606-0.8.0-Windows** and run **TRoss606-0.8.0-Windows-Setup.exe**.
2. If SmartScreen appears, click **More info → Run anyway**.
3. The VST3 goes to `C:\Program Files\Common Files\VST3`, and the app to the Start menu under Ticco Ross.

Linux

1. Unpack **TRoss606-0.8.0-Linux-x86_64.tar.gz**.
2. Run `./install.sh`. It installs for your user only, with no root needed: VST3 to `~/.vst3`, LV2 to `~/.lv2`, and the app to `~/.local/bin/TRoss606`.
3. `./uninstall.sh` removes it again.

Loading it in Ableton Live

1. Open Settings → Plug-Ins. On a Mac, turn on **Use Audio Units v2** (or **VST3**), then click **Rescan**.
2. In the browser, go to Plug-Ins → Audio Units (or VST3) → Ticco Ross → **TRoss 606** and drag it onto a MIDI track.
3. Press Play in Live: the sequencer follows Live's tempo and position.

Uninstalling on macOS: delete the three files listed in step 3 of the macOS install.

Panel tour

The panel has two halves: the silver top holds the sound, and the dark bottom holds the sequencer.



Area	Where	What it's for
Header	Top strip	The name (click it to open the About window) and, at the far right, the build version (v0.8.0)
Drum channels	Silver panel, eight columns	Tune, Decay, Level and an extra knob per drum, its name strip (click it to play the drum), M / S buttons, its trigger note and the OWN OUT switch
Master	Silver panel, far right	Accent and Volume knobs, the pattern display, and the REPEAT rate for note repeat
Step pads	Dark panel, left	9 rows × 16 steps: ACC on top, then BD, SD, LT, HT, CP, CH, OH, CY. Step numbers and the play LEDs run above them. Right-click a drum pad for its flam menu
Pattern buttons	Dark panel, right of the pads	12 slots, two across and six down. Right-click one to copy, clear or chain it; Shift-click to add it to the chain
LENGTH, RUN, CLEAR, REC	Dark panel, far right	Pattern length read-out, the sequencer keys and EXPORT MIDI. The standalone app adds PLAY and BPM under EXPORT MIDI

The sections that follow go through each area in this order.

The drum voices

Each of the 8 channels has Tune, Decay and Level, and four of them have a fourth knob that shapes their character. Hover over a knob to see its value; double-click it to go back to the default.

Knobs every channel has:

- **Tune** — pitch, −12 to +12 semitones. The clap's range is limited to one octave either way.
- **Decay** — how long the sound rings, 0–100 %.
- **Level** — channel volume, −70 to +6 dB. All the way down (−70 dB) is silent.
- **M** (mute) and **S** (solo). Solo one or more channels to hear only those. Mutes and solos fade in a few milliseconds, so they never click. The OWN OUT switch below them sends the drum to its own output (see Separate outputs).

Channel	Note	Extra knob	What the extra knob does	Defaults
Bass drum	C1	Attack	Amount of click and thud at the start of the hit	Tune +1 st, Decay 15.8 %, Attack 0 %
Snare	D1	Snappy	Level of the snare-wire noise against the drum body	Decay 75.5 %, Snappy 57.2 %
Low tom	A1	—	—	Decay 80 %
Hi tom	D2	—	—	Decay 80 %
Clap	D#1	Noise	0–50 %: noise level. Above 50 %: adds a longer, airier tail	Tune +2.6 st, Decay 100 %, Noise 37.9 %
Closed hat	F#1	—	—	Decay 70 %
Open hat	A#1	—	—	Decay 70 %
Cymbal	C#2	Tone	Brightness of the cymbal	Decay 70 %, Tone 50 %

All levels default to 0 dB and all other tunings to 0 st.

Hi-hat choke: a closed hat cuts off a ringing open hat within about 4 ms, as on the hardware. Put an open hat on step 15 and a closed hat on step 1 to hear it.

Velocity: hits up to velocity 100 scale the level (velocity 1 plays at about 30 %). Hits above 100 play at full level plus the Accent amount. Sequencer steps play at velocity 100, or 127 on accented steps.

Master section

The master section at the top right sets how hard accents hit and the overall output level, and shows which pattern is playing.

- **Accent** (0–100 %, default 50 %) — extra level for accented hits. At 50 %, an accented hit plays 1.5 times as loud as a normal one; at 100 %, twice as loud. Accents come from the ACC row of the sequencer or from keyboard hits above velocity 100.
- **Volume** (−70 to +6 dB, default −6 dB) — the output level of the whole kit.
- **Pattern display** — shows `PATTERN 3` while pattern 3 plays. When another pattern is queued it shows `PAT 3 > 5`: pattern 3 is playing, and 5 starts on the next step 1. While a chain runs it shows CHN 2/4 P1: chain step 2 of 4, pattern 1.

REPEAT (1/4, 1/8 or 1/16, default 1/16), under the MASTER strip, sets the note-repeat rate. Turn the knob or drag the read-out next to it. See *Note repeat*.

The output isn't limited. With several loud voices and accents together, peaks can go just above 0 dBFS, so leave Volume at −6 dB or add a limiter after it.

Separate outputs

Each drum can play on its own mono output, so you can give it its own effects and fader in your DAW. All drums start on the main output.

Output	Carries
Main (stereo)	Every drum whose OWN OUT is off, after Volume. The same mono mix on both channels
BD, SD, LT, HT, CP, CH, OH, CY (mono)	That drum alone while its OWN OUT is on

OWN OUT (one switch under each drum's M / S buttons, lit orange when on) moves the drum from the main mix to its own output. It doesn't play in both at once. On its own output, the drum still follows its Level, Mute, Solo and accents, but the master **Volume** doesn't affect it.

If your DAW has a drum's output switched off, that drum stays in the main mix instead of going silent.

Setting it up in Ableton Live

1. Turn on **OWN OUT** for the drum, for example the bass drum.
2. Create a new audio track.

3. In its **Audio From** menu, choose the track that holds TRoss 606. In the menu below it, choose the drum's output, for example **BD**.
4. Set the track's Monitor to **In**.
5. Repeat for every drum you want on its own track. Drums with OWN OUT off keep playing through the TRoss 606 track itself.

Other DAWs work the same way: look for the plug-in's extra outputs (often called multi-output or aux outputs) and route each one to a track or channel.

Step sequencer

The sequencer plays 16 steps of 16th notes, locked to your DAW's tempo and song position. It only runs while the DAW is playing.

The pads. Each row is one drum and each column one step. Click a pad to turn it on (its red LED lights) or off. Click and drag along a row to paint or erase several steps in one stroke. The four pad colours (cream, grey, orange, yellow) mark the four beats of the bar.

The ACC row. A pad in the top row makes every drum on that step accented. Accented steps play at full level plus the Accent amount.

Playhead. The LEDs above the grid follow the playing step, and the pads in that column get a white outline.

LENGTH (1–16, default 16) — the number of steps before the pattern starts again. Drag the red read-out up or down; double-click it to go back to 16. Pads beyond the length are dimmed but keep their notes.

RUN (default on) — turns the sequencer on or off. With RUN off, the sequencer stays silent but still counts along with the DAW, so queued pattern changes still happen on time. The keyboard still plays.

CLEAR — erases every pad in the pattern that's playing. There's no undo, so copy the pattern to a free slot first if you might want it back.

How it follows the DAW:

- Steps are counted from the start of the song. At LENGTH 16, step 1 falls on every bar line; a shorter pattern, such as 12 steps, drifts across the bar on purpose, which is useful for polyrhythms.
- Starting playback exactly on a 16th plays that step at once. Starting between steps waits for the next one.
- When Live loops or you jump to another position, the sequencer jumps with it and plays the step at the new position.

Flams

A flam adds quick, softer grace notes just before a step, like a drummer's flam. Flams are set only from the **right-click menu** of a pad, so the panel stays uncluttered. A plain click still turns the step on and off.

Right-click a drum pad (any row except ACC) for its menu:

Menu item	What it does
Step on / Step off	Turns the step on or off, the same as a click
No flam	Removes the flam from this step
Flam (1 grace note)	One grace note before the hit
Drag (2 grace notes)	Two grace notes before the hit
Ruff (3 grace notes)	Three grace notes before the hit
Spacing	How far before the step the grace notes start: 12, 25 (default), 40 or 60 ms
Grace note level	How loud the grace notes are, as a share of the main hit: 30, 50, 60 (default), 75 or 100 %
Use this flam on every hit in this row	Copies this step's flam, spacing and level to every step that's on in the row
Remove all flams in this row	Clears every flam in the row, leaving the steps on

What to know:

- Choosing a flam, spacing or level on a step that's off turns the step on first.
- A flammed pad shows one, two or three small LEDs on its left edge, one per grace note, with the main LED moved to the right. A small line under the main LED means the step has its own spacing or level instead of the defaults.
- Grace notes follow the step: they play at the step's velocity times the grace level, so an accented step gets louder grace notes too.
- Turning a step off removes its flam. Copying a pattern copies its flams, and flams are saved with your set.
- The ACC row has no flams, since it plays no sound of its own.
- Spacing is in milliseconds, so it stays the same at any tempo. The grace notes start that long before the step and are spread evenly up to it.
- On the very first step after you press Play, the grace notes would fall before the start, so that one hit plays without them.

Patterns

TRoss 606 holds 12 patterns. While the DAW plays, a new pattern always waits until the playing one reaches step 1 again, so changes land on the beat.

- **Selecting:** click a pattern button (1–12). The button you picked turns cream. Its small LED shows the pattern that's actually playing.
- **While playing:** the LED of the queued pattern blinks and the display shows `PAT 1 > 5`. When the playing pattern reaches step 1, pattern 5 takes over. Clicking the playing pattern again cancels the queued change.
- **While stopped:** the change is immediate.
- **Editing:** the pads always show and edit the pattern that's playing, not the queued one. To edit a pattern, select it and let it start, or stop the DAW first.
- **LENGTH, RUN and the sounds are shared** by all 12 patterns. The pads, including the ACC row, are stored per pattern.

Automating pattern changes: the Pattern selector is an automatable parameter. Draw automation for it in Live's arrangement (or use clip envelopes) to change patterns along the song. The queueing rule still applies, so an automation point just before a bar line switches cleanly on that bar. While a chain is set, the chain decides which pattern plays and Pattern automation is ignored.

Copying patterns

Right-click a pattern button for its menu:

- **Copy to** → choose another pattern. Every pad of this pattern, ACC row included, is copied over the one you choose, which is replaced completely. There's no undo, so check the target first.
- **Clear** empties the pattern (greyed out when it's already empty).

Empty patterns show a small dash in the bottom-right corner of their button, so you can see at a glance which slots are free.

Chaining patterns

A chain plays several patterns one after another, for example a groove three times and then a fill.

1. **Shift-click** pattern buttons in the order you want them played. Each Shift-click adds one step to the chain, up to 32 steps, and the same pattern can appear more than once. For a groove three times and a fill: Shift-click 1, 1, 1, 2.
2. Press Play. Each pattern plays its full LENGTH, then the next one starts on step 1. At the end, the chain starts again.

What to know:

- **Locked to the song position.** Bar 1 of the song plays chain step 1, bar 2 plays step 2, and so on, a "bar" being one pattern LENGTH. Jumping, looping or starting in the middle of the song always lands on the right pattern.
- **Lights and display.** Chained buttons show a dim LED, and the one playing is bright. The display shows `CHN 2/4 P1`: chain step 2 of 4, playing pattern 1.
- **Editing.** The pads show and edit the pattern that's playing in the chain, and REC records into it.
- **Ending the chain.** A plain click on a pattern button ends the chain and goes back to that single pattern.
- **Right-click menu. Add pattern N to the chain**, **Remove the last chain step** and **Clear the chain**.
- **Saving and export.** The chain is saved with your Live set, and EXPORT MIDI can write it as one clip (*The chain, in a row*).

Exporting MIDI

The **EXPORT MIDI** key, under REC, turns your patterns into standard MIDI files that any DAW can open.

Click it and a menu opens. The menu's title shows the LENGTH and tempo that will be used.

Menu item	What you get
Pattern 3 (playing)	One file with the pattern that's playing. Greyed out when that pattern is empty
All patterns, in a row	One file with every pattern that has notes, one after another (1, 2, 3…). Empty patterns are skipped
All patterns as separate files	One file per pattern with notes, named `TRoss606 Pattern 1.mid`, `TRoss606 Pattern 2.mid`…, in a folder you choose
The chain, in a row	One file with the chain played through once, pattern after pattern. Only when a chain is set

A save dialog follows with a ready-made file name. Files start in your Documents folder.

Drag it onto a MIDI track in Live (or another DAW that accepts dropped files) and the playing pattern lands there as a MIDI clip, with no save dialog. Drag at least a few pixels so it isn't taken as a click.

What the file contains

Item	How it's written
Timing	One step = one 16th note. Each note lasts just under a 16th
Length	The pattern's LENGTH. A 12-step pattern exports 12 steps, not a whole bar
Notes	TRoss 606's own map: BD C1, SD D1, LT A1, HT D2, CP D#1, CH F#1, OH A#1, CY C#2
Velocity	100 on normal steps, 127 on accented steps (the ACC row)
Tempo	The song's tempo when you export (120 BPM if the song has never played)
Channel	10, the standard drum channel
Flams	Grace notes are written as short, softer notes just before the step, with the step's spacing (at the song's tempo) and grace level

Because the notes and velocities match TRoss 606's own map, an exported clip played back into TRoss 606 sounds exactly like the pattern, accents included. It also plays correctly on a Drum Rack or any General MIDI drum kit.

The export writes the programmed notes as they are: Mute and Solo don't remove a drum from the file. Live may round a short clip, such as 12 steps, up to a full bar; set the clip's loop length back to 12 steps if you need it.

Playing and recording drums

Any MIDI note on the track plays a drum, whether the DAW is playing or stopped. The map follows the General MIDI drum layout, with Ableton's note names (C1 = MIDI note 36). Notes not listed are ignored, and note-offs do nothing.

Drum	Main note	Also plays on
Bass drum	C1 (36)	B0 (35)
Snare	D1 (38)	E1 (40)
Low tom	A1 (45)	F1 (41), G1 (43)
Hi tom	D2 (50)	B1 (47), C2 (48)
Clap	D#1 (39)	—
Closed hat	F#1 (42)	G#1 (44)
Open hat	A#1 (46)	—
Cymbal	C#2 (49)	D#2 (51), E2 (52), G2 (55), A2 (57)

This means MIDI clips made for Live's Drum Rack or other GM kits play on TRoss 606 without changes. Velocity works as in **The drum voices**: up to 100 sets the level, and above 100 adds the accent.

Playing from the panel

Each drum's black name strip (BASS DRUM, SNARE and so on) works as a pad. Click it and the drum plays straight away, whether the DAW is playing or stopped. The strip flashes orange on each hit.

- **Click:** a normal hit, the same as velocity 100 from a keyboard.
- **Shift-click:** an accented hit, the same as velocity 127. The strip flashes white.

With REC on and the DAW playing, clicks are recorded exactly like keyboard hits (below). With REC off, or the DAW stopped, a click only plays the sound and never changes the pattern.

Note repeat

Hold Option (Alt on Windows) and press a drum's name strip. The drum keeps hitting at the **REPEAT** rate for as long as you hold the mouse button down, and stops when you let go. The strip flashes on every hit.

REPEAT	Hits	At 120 BPM
1/4	One per beat: steps 1, 5, 9, 13	2 per second
1/8	Every other step: 1, 3, 5…	4 per second

REPEAT	Hits	At 120 BPM
1/16	Every step	8 per second

- **While the song plays,** the first hit sounds the moment you press. The repeats then land exactly on the song's 1/4, 1/8 or 1/16 grid, so they're always in time.
- **Recording:** with REC on and the song playing, every repeat is written into its step. That's a quick way to fill a hi-hat line: hold Option on CLOSED HAT for one bar at 1/16.
- **Accent:** hold Shift as well (Shift + Option) for accented repeats. With REC on they also set the ACC pads.
- **While stopped,** the drum repeats at the REPEAT rate counted from your press, at the song's tempo, and nothing is recorded.
- Only one drum repeats at a time. Pressing another name strip with Option switches the repeat to that drum.

Recording into the pattern

With **REC** on (the default, lit orange) and the DAW playing, every drum you hit is written into the playing pattern:

1. Start playback. The sequencer runs as usual.
2. Play drums on your keyboard, or click their name strips on the panel. Each hit goes into the **nearest** step, so a note played slightly early lands on the step coming up, and one played slightly late on the step that just passed.
3. Hit harder than velocity 100, or Shift-click a name strip, to set the ACC pad on that step as well.
4. Recording only adds notes. To remove one, click its pad, or use CLEAR for the whole pattern.

Turn **REC** off to play along without changing the pattern. Recording never happens while the DAW is stopped.

Saving and automation

Everything is saved with your Live set (or your DAW project): every knob and button, all 12 patterns, and which pattern was playing. There's no separate preset file. To keep a kit, save the track as a Live preset or in a set you reuse.

All 59 parameters can be automated or mapped to a MIDI controller. The grid pads aren't parameters, so they can't be automated.

Parameters	Range	Default
BD / SD / LT / HT / CP / CH / OH / CY Tune	−12 to +12 st	0 st (BD +1 st, CP +2.6 st)
BD / SD / LT / HT / CP Decay	0–100 %	80 % (BD 15.8 %, SD 75.5 %, CP 100 %)
CH / OH / CY Decay	0–100 %	70 %
BD / SD / LT / HT / CP / CH / OH / CY Level	−70 to +6 dB	0 dB
BD Attack	0–100 %	0 %
SD Snappy	0–100 %	57.2 %
CP Noise	0–100 %	37.9 %
CY Tone	0–100 %	50 %
BD … CY Mute, BD … CY Solo, BD … CY Own Out	off / on	off
Accent	0–100 %	50 %
Volume	−70 to +6 dB	−6 dB
Seq On (RUN)	off / on	on
Seq Length	1–16	16
Pattern	1–12	1
Record (REC)	off / on	on
Repeat Rate (REPEAT)	1/4, 1/8, 1/16	1/16

The parameter names match the Max for Live device, so automation is easy to recreate if you move a set over from the device.

Standalone app

The standalone app runs TRoss 606 without a DAW. Since there's no host transport, it adds two controls at the bottom right, under EXPORT MIDI:

- **PLAY** — starts and stops the sequencer from step 1.
- **BPM** — tempo, 40–240. Drag the red read-out up or down.

Everything else works as in the plug-in, including recording from a MIDI keyboard while PLAY is on. The app has only the main output, so it doesn't show the OWN OUT switches.

To choose the audio output and the MIDI input, use the app's **Options** button (top left of its window) → **Audio/MIDI Settings**. Pick your audio device, then tick your keyboard under *Active MIDI inputs*.

When you quit, the app saves its last state (knobs and all 12 patterns) and brings it back next time. It keeps only one state, though, so use the plug-in inside a DAW for kits and patterns you want to keep separately.

About and the built-in manual

Click the TRoss 606 logo at the top left of the panel to open the About window. Click anywhere in it, or press Esc, to close it.

Item	What it shows
Version	The build version, for example v0.8.0
Built	The date this build was made
Running as	The plug-in format (AU, VST3, LV2 or Standalone) and, when it can tell, the host
Contact	ticcoross@gmail.com. Click the address to write an email
Open the manual	Opens this manual as a PDF
Quick reference	The main mouse and key shortcuts on one page

The manual is built into the plug-in, so **OPEN THE MANUAL** works offline and always matches the version you're running. It's written to your computer's temporary folder and opened in your PDF viewer.

Tips and troubleshooting

Tips

- **Classic 606 bass drum:** the defaults (Tune +1 st, Decay 15.8 %, Attack 0 %) give a short, tight kick. Raise Decay above 70 % for a long, booming one.
- **Open hat with choke:** program open hats on the off-beats and closed hats on the beats. Each closed hat cuts the open hat short, as on the hardware.

- **Accent grooves:** put ACC on steps 1, 4, 7, 11 and 15 and keep the drums the same. One row changes the whole feel.
- **Polyrhythms:** set LENGTH to 12 or 7 and let it run against your 16-step clips in Live.
- **Separate processing:** turn on OWN OUT for the kick and snare and route them to their own tracks for compression or reverb. The hats and toms can stay in the main mix.
- **Window size:** drag the bottom-right corner to resize the window, from 75 % to 200 %.

Troubleshooting

Problem	What to do
macOS won't open the installer	System Settings → Privacy & Security → Open Anyway. The installer isn't signed.
TRoss 606 isn't in Live's browser	Settings → Plug-Ins: turn on *Use Audio Units v2* or *VST3*, then hold Option (Mac) or Alt (Windows) and click Rescan.
The sequencer doesn't play	Check that the DAW is playing and RUN is lit. Check that the pattern's pads aren't all off, and that no channel is muted or soloed by mistake.
A pattern change doesn't happen	It's queued until step 1 (its LED blinks). With a long LENGTH, wait for the pattern to wrap.
A recorded hit plays twice	You hit slightly before a step, so the hit was written onto the step coming up, which then plays it again. That's normal; the next pass plays it once.
Notes from the keyboard aren't recorded	REC must be lit, and the DAW must be playing. Also check that the notes are in the map in *Playing and recording drums*.
Output clips	Lower Volume or a few channel Levels, or reduce Accent.

Specifications

- Version 0.8.0, built with JUCE 8.0.9.
- Sample rates: any the host uses; tested at 44.1, 48 and 96 kHz.
- Plug-in IDs: manufacturer `Tcro`, plug-in `T606` (AU type `aumu`).
- Source and builds: github.com/Spunkmeyer77/TRoss606.

Credits

TRoss 606 by Ticco Ross. Drum voices based on 606-Inspired-Synth-Drums by Matthew Fecher, MIT licence. TR-606 is a trademark of Roland Corporation; TRoss 606 is not affiliated with Roland.